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Polymetallic Seafloor Minerals in the Indian Ocean

Executive Summary: The Indian Ocean hosts extensive deep-sea mineral resources, notably polymetallic *manganese nodules* on abyssal plains (especially in the Central Indian Ocean Basin, CIOB), *ferromanganese crusts* on seamounts and ridges, and *hydrothermal polymetallic sulphides* along mid-ocean ridges. The Indian government leads exploration in this region: India holds ISA exploration contracts for nodules (CIOB) and two sulphide areas (Central Indian and Carlsberg ridges)【9†L288-L292】【30†L53-L60】. India's nodules concession (75,000 km²) is estimated to contain ~380 Mt of nodules with ~4.7 Mt Ni, 4.29 Mt Cu, 0.55 Mt Co and 92.6 Mt Mn【47†L18-L24】. Cobalt crusts (on volcanic seamounts) are known worldwide but Indian Ocean resource estimates are unreported; crusts are rich in Co (up to ~2%) and rare earths【67†L206-L209】. Hydrothermal sulphides (Fe–Cu–Zn–Au–Ag–Pt deposits) occur on slow-spreading ridges (e.g. Central and Southwest Indian Ridges, Carlsberg Ridge); India's ISA contracts cover ~15,000 km² (10,000 km² on Carlsberg)【30†L35-L42】【12†L276-L278】. Extraction technologies (e.g. deep-sea collectors, riser pumps, processing systems) are under development. Environmental studies (e.g. India's INDEX experiment) highlight impacts on benthic communities (unique fauna on nodules/crusts) and large sediment plumes. The legal framework is UNCLOS/ISA-regulated "Area" mining and national EEZ rules for inshore waters. Commercial viability remains uncertain: projected CAPEX runs into tens of billions (one company estimated ~$38B through 2046【88†L168-L172】), and profitability depends on volatile Ni/Co/REE prices and demand. Key stakeholders include India's Ministry of Earth Sciences (NCPOR), international ISA contractors (e.g. China's COMRA, Korea), deep-sea mining firms, and research bodies (e.g. IFREMER, USGS). Major data gaps remain (e.g. detailed reserve estimates for crusts/sulphides in the Indian Ocean, long-term ecosystem impacts) and ongoing research is recommended.

Geological and Tectonic Setting

The Indian Ocean formed by Gondwana breakup; it is bounded by Asia, Africa and Australia (Figure: Indian Ocean region). Major tectonic features include the mid-ocean ridges (Central and Southwest Indian Ridges, Carlsberg Ridge) and fracture zones (Owen Fracture Zone, Ninety East Ridge). Oceanic crust is typically basaltic and relatively young (e.g. < 120 Ma). Abyssal plains (4000–6000 m depth) dominate most of the central basins (CIOB, Southern Indian Basin), while ridges and seamount chains rise up to ~1000–3000 m below sea level. These topographic settings control mineral deposition: nodules form on flat abyssal clay plains, crusts on seamount flanks, and sulphides at hydrothermal vents along ridges【2†L221-L224】【67†L198-L204】. Sediment accumulation is very low in the central basins (only a few mm/ky), allowing minerals to accumulate over millions of years.

Bathymetry and Sedimentology

The ocean floor depths average ~3741 m【58†L394-L402】. Abyssal plains in the CIOB lie mostly between 4500–5500 m, with slow sedimentation (clays, siliceous oozes). Manganese nodules require extremely low sedimentation so that metal precipitates (from seawater or pore fluids) build on hard nuclei【2†L211-L219】. Sediment cores from the CIOB show a complex mix of biogenic (siliceous), volcanic and clastic layers, but no thick turbidites near the mid-ocean basins. Seamounts (e.g. volcanic knolls rising thousands of meters) break the sediment cover and support ferromanganese crusts; their

slopes trap hemipelagic sediments and nodules on the flat summits. Hydrothermal vents occur on ridge flanks where sediment cover is thin or absent.

Polymetallic Nodules

Occurrence: The main nodule province in the Indian Ocean is the Central Indian Ocean Basin (CIOB) (roughly 10–20°S, 70–80°E). India's ISA contract area (~75,000 km^2) covers much of this. No significant nodule fields are known in EEZs of littoral states. Index sampling (via ROVs and dredges) has found nodules ~2–15 cm diameter spread over abyssal clays. Nodules are hydrogenous/diagenetic precipitates of Mn, Fe and trace metals.

Resource Estimates: A 2017 Indian government report estimates ~380 million tonnes of nodules in the CIOB contract area【47†L18-L24】. This implies grades of ~1.24% Ni, 1.13% Cu, 0.15% Co and 24.4% Mn by weight (based on 4.7 Mt Ni, 4.29 Mt Cu, 0.55 Mt Co, 92.59 Mt Mn in 380 Mt nodules)【47†L18-L24】. Sand and other oxides make up the balance. By comparison, economic recovery targets typically require combined Ni+Cu+Co >2%【2†L221-L224】. India's long-term mine-site planning has identified ~7,860 km^2 of high-grade fields within the IOC (the "first-generation mine site")【47†L35-L39】. Other estimates outside the Indian Ocean (e.g. CCZ) are much larger, but the CIOB remains one of the world's largest single nodule resources.

Distribution and Grades: Nodule coverage in the CIOB varies (often ~1–15 kg/m^2). Areas exceeding 5 kg/m^2 are considered potentially economic【2†L221-L224】. Metal contents vary laterally; Indian studies note Mn is highest where Co is modest, reflecting chemical zoning during growth【47†L18-L24】. Small (<5 cm) nodules dominate most areas, with fewer very large ones compared to Pacific fields. No publicly accessible dataset exists for fine-scale nodules distribution; general bathymetric/sediment maps (e.g. GEBCO) indicate relatively flat abyssal plain depths of ~5000 m across most of the field.

Technology and Demonstrations: India has developed deep-sea mining technology under its Deep Ocean Mission. Notably, a **Remotely Operated Submersible (ROSUB 6000)** was built/tested to 5289 m【47†L41-L45】. A seafloor "bucket-ladder" style *mining system* was tested in shallow water (500 m) to validate excavation and pumping concepts【47†L45-L49】. A 500 kg/day pilot plant for metallurgical processing (Ni/Cu/Co extraction) operates in India【47†L45-L49】. Collector designs (generally tracked or towed vehicles with sediment entrainment pumps) and slurry risers to surface ships are standard concepts, but no full-scale commercial system exists yet.

Environmental Baseline: The nodule field hosts diverse abyssal communities. Studies in the Pacific show nodules provide critical hard-substrate habitat for sponges, corals and other fauna; non-nodule areas have distinctly different communities. Indian surveys (part of INDEX) recorded deep-sea fish, crustaceans, and sessile fauna on nodules. The environment is very oligotrophic. Baseline oxygen levels are moderate (>3 ml/L) but the ecosystem is slow to recover from disturbance (years to centuries). Nodule removal would eliminate hard habitat; sediment plumes could smother nearby fauna. India's index dredging experiment and ISA REMP workshops highlight low bioturbation rates and unique taxonomy in the CIOB, suggesting high sensitivity【2†L221-L224】.

Cobalt-Rich Ferromanganese Crusts

Occurrence: Crusts form on hard rock (seamounts, ridges) between ~400 and 5000 m【67†L198-L204】. In the Indian Ocean, candidate sites include submarine plateaus off Madagascar and the

Mascarene Plateau, as well as seamounts along the Carlsberg and Central Indian ridges. However, no ISA exploration contracts currently target Indian Ocean crusts; known contracts for crusts are in Pacific and Atlantic【67†L219-L227】.

Composition: Crusts are metal-rich surfaces up to 20 cm thick. They contain Fe, Mn, Co (often 0.5–2%), Ni (0.5–1.5%), Cu, along with rare earths and platinum-group elements【67†L198-L204】【67†L206-L209】. The highest grades are on steep seamount slopes where sediment does not cover the substrate.

Resource Estimates: No published tonnage or grade estimates exist for Indian Ocean crusts. For context, global crust tonnage (Pacific prime zone) is on the order of 7.5 billion dry tons【76†L26-L33】, but Indian Ocean amounts are unreported. Given India's extensive seafloor in this region, substantial crust resources likely exist, but detailed mapping and sampling are needed. Pending exploration, we mark crust reserves in the Indian Ocean as "unspecified".

Extraction Technology: Crust mining would likely use remotely-operated drill/cutting vehicles or large underwater scrapers to remove crust layers (they adhere to rock by tens of millions of years of growth). Proposed methods include tracked excavators or high-pressure water jets to detach crusts, then pumping slurry to surface. No full-scale prototype has been built yet. Crust operations need precise targeting of topography highs and containment of fine sediment.

Environmental Impacts: Removal of crust would destroy hard-rock habitats. Seamounts often harbor rich communities (corals, sponges) that are poorly known in the Indian Ocean. Disturbance would leave rock surfaces bare and generate sediment plumes. Because crusts grow ~mm/ka, recovery is effectively impossible. Environmental impact assessments would be required, but almost nothing is publicly known for Indian seamount ecosystems; ISA REMP guidelines note seamounts as vulnerable【67†L198-L204】.

Polymetallic Sulphides

Occurrence: Hydrothermal vent fields occur along mid-ocean ridges. In the Indian Ocean these include vents at the Carlsberg Ridge, Central Indian Ridge (e.g. Edmond, Kairei fields), and Southwest Indian Ridge. India's ISA contract (signed 2025) allocates 10,000 km² on the Carlsberg Ridge【30†L35-L39】, and an earlier contract (2016) covers Central Indian Ridge areas【30†L59-L64】. China (COMRA) holds an ISA area on the Southwest Indian Ridge【12†L254-L260】; Korea and Germany also have ISA sulphide contracts on Central Indian Ridges【12†L258-L263】. In general, these are deep (2500–3500 m) and remote volcanic terrain, with active vents producing metal-laden chimneys.

Composition: Sulphide deposits (seafloor massive sulphides, SMS) are rich in Fe, Cu, Zn, with gold, silver and platinum-group elements. One Indian report notes "valuable metals such as iron, copper, zinc, silver, gold, and platinum" in PMS【30†L53-L57】. Grades can vary but typical SMS cores can exceed 20% Cu+Zn. Rare Earths are generally low in vent sulphides. The presence of precious metals is one reason PMS are economically interesting.

Resource Estimates: No large-scale tonnage estimates exist for Indian Ocean SMS. Individual vent fields are small (typically <1 Mt) and discontinuous. The ISA contracts (e.g. India's 10,000 km² Carlsberg zone) likely contain multiple vent prospects, but published resource tonnages or metal content are not available. We therefore list sulphide resources as "unspecified" for lack of public data. (By contrast, mid-ocean ridge deposits globally are modest in size; even large SMS deposits are orders of magnitude smaller than nodules or crusts.)

Mining Technology: Proposed methods for sulphides differ from nodules. Concepts include remotely operated mining machines ("crawler" or cone cutters) that chip sulphide ore and pump slurry to surface. Alternatively, riser-less pumping of dry ore using gravity (a "crawler robot" dumping into a submersible container lifted by a cable) has been suggested. No firm technology is mature; vent fields often lie on slope terrain. Collectors must avoid clogging on sandy soils. Processing may involve on-board separation or bringing slurries to shore for refining.

Environmental Impacts: Hydrothermal vent ecosystems are highly endemic (unique tube worms, shrimps, microbes). Mining risks include complete destruction of vent communities and unknown impacts on ocean chemistry (metal dispersal). Plume simulations at Pacific sites suggest sediment may settle over hundreds of meters, potentially smothering background fauna【19†L213-L219】. In Indian context, little is known about vent biota; ISA guidelines note vent fields as conservation priorities. Clearance thresholds and baseline studies are lacking in the Indian Ocean, a major knowledge gap.

Exploration Licences and Regulatory Status

All deep-sea exploration in international waters (the "Area" under UNCLOS) is regulated by the **International Seabed Authority (ISA)**. In the Indian Ocean context, the ISA has issued:

- **Polymetallic Nodule**: India (Govt. of India, via NCPOR) holds a contract (since 2002, renewed) covering ~75,000 km^2 of the CIOB【47†L18-L24】【9†L288-L292】. The first 15-year period (2002–2017) was extended by 5 years in 2017【47†L18-L24】, and a second period is now ongoing (to 2027). No other nation has an ISA nodule license in the Indian Ocean; Indian nodules are the sole focus here.

- **Polymetallic Sulphides**: ISA contracts include China (COMRA) for Southwest Indian Ridge, Korea for Central Indian Ridge, Germany (BGR) for Central Indian Ridge, and **India for two areas**: one on the Central Indian Ridge (contract since 2016) and a second on the Carlsberg Ridge (contract signed Sep 2025 for 10,000 km^2)【30†L35-L39】【12†L276-L278】. India is thus the first country with two ISA sulphide contracts【30†L59-L64】. Contract durations are 15 years (renewable) – e.g. India's Carlsberg contract runs 2025–2040.

- **Cobalt Crusts**: No ISA contract has been allocated yet in Indian Ocean waters; existing crust exploration is in the Pacific/Atlantic【67†L219-L227】. However, many crust deposits lie within EEZs of coastal states (Mauritius, India, etc.) and would be subject to national jurisdiction if pursued.

Within **EEZs (national waters)**, some coastal states permit seabed exploration under national law. India's EEZ has been surveyed for minerals by national agencies, but no commercial licences have been granted. Countries like India and France (Reunion/Mayotte EEZ) maintain regulations for marine minerals, but little published information exists. All coastal states are bound by UNCLOS to manage resources sustainably.

A summary table of known contracts is given below:

Contractor / Agency	Sponsoring State(s)	Resource Type	Region / Location	Area (km²)	Contract Period
Government of India (NCPOR)	India	Polymetallic nodules	Central Indian Ocean Basin	~75,000 【47†L18-L24】	2002–2022 (extended to 2027)【47†L18-L24】【9†L288-L292】
Government of India (NCPOR)	India	Polymetallic sulphides	Central Indian Ridge	(unspecified)	2016–2031 【12†L270-L272】
Government of India (NCPOR)	India	Polymetallic sulphides	Carlsberg Ridge	~10,000 【30†L35-L39】	2025–2040 【12†L276-L278】
China Ocean Mineral R&D Assoc.	China	Polymetallic sulphides	Southwest Indian Ridge	(unspecified)	2011–2026 【12†L254-L260】
Government of Korea	Korea	Polymetallic sulphides	Central Indian Ridge	(unspecified)	2017–2032
BGR (Germany)	Germany	Polymetallic sulphides	Central Indian Ridge (“CIO”)	(unspecified)	2015–2030
EEZ (national waters)	*Various*	Nodules/ Crusts/ Sulphides	Indian Ocean EEZs	NA	National laws apply

Table: Key exploration contracts in the Indian Ocean【47†L18-L24】【12†L270-L278】.

Mining and Extraction Technologies

No commercial deep-sea mining has yet occurred; technologies remain in pilot stage. For **nodules**, the envisioned approach is a *seafloor collector* (tracked vehicle or towed continuous-bucket system) that either scoops or pumps sediment/nodules, with an onboard pump transferring slurry up a riser to a surface ship or processing vessel. India’s ROSUB 6000 and bucket tests exemplify research efforts 【47†L41-L49】. Riser technology must handle abrasive, dense slurry at 4–5 km depth – designs mirror oilwell drilling risers but must avoid clogging (studies are ongoing). Processing options include shipboard separation or dewatering, or pumping slurry ashore.

For **hydrothermal sulphides**, proposals include either a similar slurry-pump approach (pump crushed ore via riser) or *riser-less* “bulk transport” vehicles that collect ore in subsea containers periodically lifted to the surface. The latter avoids long slurry columns but requires heavy lift capability. Because sulphides lie on slope terrain, mining machines need good traction and stability. For **crusts**, an undersea bulldozer or high-pressure water cutter might scrape crust from rock, with a pump hauling the cut material; hard-rock drilling has also been proposed. In all cases, energy (electric/hybrid ships), automation (ROVs or autonomous vehicles), and cold/wet metallurgy are challenges. Several companies

(DeepGreen/The Metals Company, Nautilus Minerals (defunct), German BGR, etc.) have prototype designs in development. India's contracts imply NCPOR/CSIR and industry partners are also designing equipment under the Deep Ocean Mission, but most details remain confidential.

Environmental Impacts and Ecosystems

Deep-sea mining poses novel risks. **Benthic communities** in the CIOB are unique: nodules host brittle stars, sponges, amphipods and rare corals; crust-covered seamounts shelter gorgonians and fish. Disturbance (e.g. from dredging) kills organisms directly and creates sediment plumes. Experiments show plumes may travel kilometres laterally【19†L219-L227】 and settle on passive suspension feeders, clogging gills and smothering larvae. Mining also removes substrata (nodules/crusts) that took millennia to accrete. Recovery times are predicted in centuries or longer.

Baseline surveys in the Indian Ocean are sparse. The ISA's "INDEX" program (1980s–2000s) conducted preliminary benthic studies in the CIOB, noting low biomass and diversity, but also some endemic species. A 2023 ISA-India REMP workshop identified large data gaps in connectivity and planktonic larval dispersal in this region. Deep Sea faunal studies emphasize caution: communities adapted to stable, oligotrophic conditions may not rebound quickly from disturbance. Additionally, potential pollution (oil leaks, noise) and microbiological impacts (redistribution of heavy metals, alteration of element cycles) are unknown. Regional environmental management plans (REMPs) are under development; Indonesia and Mauritius have begun marine protected area planning in their EEZs. To date, no ecosystem in the Indian Ocean has been studied intensively enough for definitive risk assessment.

Regulatory and Legal Framework

All activities in the Area (high seas seabed) fall under UNCLOS and ISA's Mining Code. India ratified UNCLOS and is a pioneer investor; its Ministry of Earth Sciences/NCPOR implements contracts. Contractors must follow the ISA's Exploration and (draft) Exploitation Regulations, including environmental rules (pollution prevention, thresholds, contingency plans). They must also train personnel from developing States and submit annual reports. Coastal States regulate their EEZ minerals through national laws (e.g. India's Marine Mineral regulations), but no Indian Ocean state has yet granted a mining lease.

Internationally, tensions exist: ISA is still finalizing exploitation rules, with some countries (and NGOs) calling for moratoria. India actively participates in ISA Council and LTC deliberations. As noted, TMC (DeepGreen) has sought to sidestep ISA by partnering with a non-UNCLOS state, highlighting geopolitical complexity. Any venture must also comply with biodiversity treaties (e.g. CBD, regional seas conventions) and Nationally Determined Contributions (if deep-sea ecosystems become part of climate/biodiversity commitments). Current policy trends emphasise careful science-based management (e.g. the IOC-UNESCO/ISA International Guidelines).

Commercial Viability and Cost Drivers

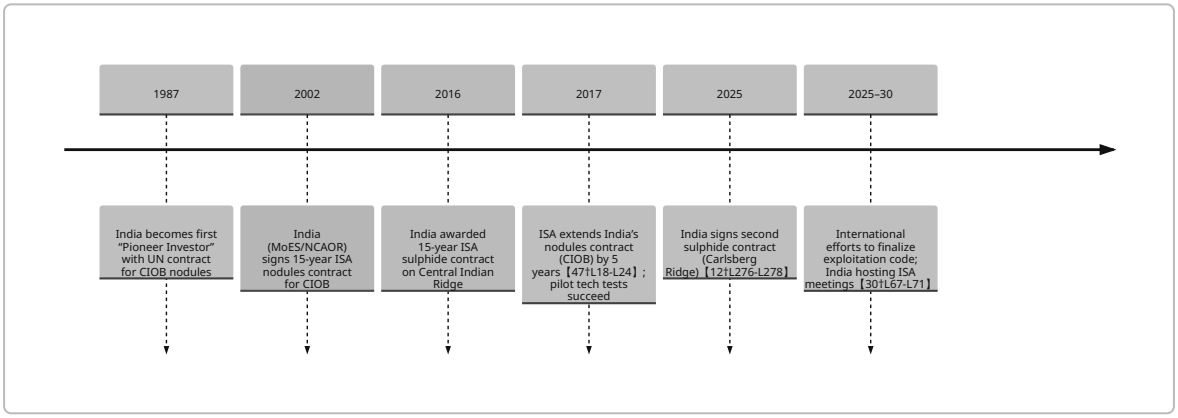
No deep-sea mine is yet operating, so cost estimates are speculative. Early studies suggest very high CAPEX: one estimate based on a nodules project was ≈US$38 billion total (by 2046)【88†L168-L172】, not accounting for inflation or interest rates. Operating costs are also high due to remote operations and energy demands. Key sensitivities include metal prices (particularly Ni and Co). With global shifts (e.g. EV batteries moving away from Ni/Co) and current oversupply of these metals【88†L128-L132】, future revenues are uncertain.

On the revenue side, nodules contain many tons of Ni/Cu/Co, potentially displacing terrestrial mining. The 380 Mt CIOB resource could yield on the order of 5–6 million tonnes of Ni+Cu combined. If commodity prices remain as in 2026 (around Ni US$20/kg, Cu US$10/kg), gross revenue might approach tens of billions. However, environmental mitigation and regulatory costs could be substantial. For crusts, high cobalt content is attractive, but crust mining yields relatively low tonnages (metals are dispersed). Sulphides' value lies in multiple metals plus precious elements; a hypothetical Carlsberg mine could produce several thousand tonnes of Cu/yr and a few hundred kg Au/yr, but requires detailed feasibility which is lacking.

Overall, analyses note that deep-sea mining profitability is marginal without strong price assumptions. For example, one financial review concluded that even large nodule projects would only supply a few percent of Ni/Co market by 2025【88†L160-L168】. A rigorous cost-benefit assessment (particularly in 2026/27) is essential; at present, capital costs and permitting timelines (likely >10 years) mean any commercial venture is a long-term play.

Timelines and Stakeholders

Deep-sea mineral development in the Indian Ocean is unfolding over decades. Key milestones include:



Key stakeholders in the Indian Ocean include: India's **Ministry of Earth Sciences / NCPOR**, which leads national exploration; scientific institutions (CSIR-NIO, NIOT); ISA and its Legal/Technical Commission; and academic partners (e.g. Marine Institutes of India, Japan's JAMSTEC). Active companies are mostly international (e.g. DeepGreen/TMC, Nautilus-affiliated entities, Ocean Minerals Japan), though many have global projects. China's COMRA, Korea's KIGAM, BGR (Germany), and IFREMER (France) are involved regionally via ISA. NGOs and scientific consortia (e.g. ISA-UNEP workshops) also influence policies.

Uncertainties and Research Needs

Data Gaps: Detailed maps of nodule abundance and crust occurrences in the Indian Ocean remain incomplete. Biogeographic baselines for deep-sea species are sparse. Technology readiness (for 6 km depth) is unproven. The economic models rely on future prices that are highly uncertain.

Future Directions: Recommended research includes targeted multibeam surveys to refine resource mapping; long-term benthic monitoring at potential mine sites; hydrodynamic modelling of mining plumes; and pilot field experiments under controlled conditions. Cooperative REMP development under ISA and national policies will be critical. Most importantly, transparent peer-reviewed studies on costs

and market models are needed to inform whether Indian Ocean deep-sea mining could ever be commercially viable.

This report is compiled from open-source literature, agency reports and ISA documentation. Key figures (map of Indian Ocean seabed tectonics, regional mineral occurrences) and tables (resource estimates, license lists) should be consulted for detailed planning. (Areas lacking public data are marked "unspecified" above.)
